

PAPER



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Unveiling the electronic and magnetic landscape of 3d transition metal doped hydrogenated borophenes: a first-principles study†

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In this study, density functional theory (DFT) simulations have been utilized to probe the intricate electronic and magnetic properties of pristine and 3d transition metal doped hydrogenated borophenes. It has been investigated through electronic structure calculations that the hydrogenation of 2-*Pmmn* borophene leads to the emergence of an in-plane Dirac cone, elucidating its transformation into a potential Dirac material with fortified stability. By employing spin-polarized DFT calculations with the Hubbard *U* correction, we have estimated the electronic and magnetic states of transition metal doped hydrogenated borophenes. Our analysis reveals that the Cr doped hydrogenated borophene manifests the highest magnetic moment of $4.76\mu_B$, making it a promising magnetic 2D material. Furthermore, the exchange energy has been calculated by considering the interaction between two transition metal atoms, to assess its magnetic state (ferromagnetic/antiferromagnetic/non-magnetic). The mean field theory and Heisenberg model have been utilized for Néel and Curie temperature estimation, corresponding to anti-ferromagnetic and ferromagnetic states respectively. The present study contributes to the design and understanding of Dirac materials with tailored electronic and magnetic characteristics, highlighting the potential for novel applications in electronics and spintronics. The insights gained from this work may pave the way for future experimental investigations and the realization of functionalized 2D materials with tunable properties.

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1. Introduction

The discovery of two-dimensional (2D) materials after the huge success of graphene in 2004 by Novoselov *et al.*¹ has revolutionized the field of materials science related to 2D materials such as silicene,² germanene,³ phosphorene,⁴ stanine,⁵ borophenes,⁶ *etc.* These 2D materials have been widely studied, as they show unique optical, mechanical, magnetic and electronic properties, due to their quantum confinement effect and other exotic 2D features.⁷

Amongst these 2D materials, borophene, a single layer of boron atoms, has gained significant attention due to its proximity to carbon in the periodic table and having a non-planar anisotropic structure. Liu *et al.*,⁸ by using first-principles calculations, predicted that boron sheets can be grown on the Ag(111) or Au(111) surfaces by deposition. The experimental realization of buckled 2-*Pmmn* borophene on a silver Ag(111) surface has been achieved for the first time by Mannix *et al.*⁹ under ultrahigh-vacuum conditions. As borophene is inherently metastable, other

phases of borophene, like β_{12} , χ_3 , single crystal and honeycomb borophene sheets, have also been grown experimentally on metal substrates.^{10–12} Amongst all the phases of borophene, the buckled triangular phase (2-*Pmmn*) is highly anisotropic and shows novel mechanical properties, *e.g.* an enhanced in-plane Young's modulus (second-order elastic stiffness) value of 398 N m^{-1} along the un-corrugated direction,⁹ which is even higher than that of graphene (340 N m^{-1}).¹³ Borophene exhibits significant potential for applications in metal-ion batteries as an anode material due to its high capacity, exceptional electronic and ionic conductivity, *etc.*^{14,15} In addition to its anisotropic and metallic properties, borophene exhibits superconductivity with a superconducting transition temperature (T_C) in the range of 10–20 K.¹⁶ Borophene tends to get oxidized outside the ultrahigh-vacuum chamber as a consequence of its highly reactive surface, which hinders the practical usage of borophene in its pristine form. In addition to this, the current challenge for borophene is to develop it with dynamic stability in a free standing state.⁹

One of the effective ways to stabilize borophene is by surface functionalization, *e.g.* hydrogenation,¹⁷ fluorination,¹⁸ *etc.* The structural, electronic and magnetic properties of functionalized graphene,¹⁹ silicene, germanene, stanine, plumbene,²⁰ *etc.* have been studied in detail. The surface functionalization of

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borophene by hydrogenation is a prominent approach as it increases the dynamic stability of borophene and exhibits a Dirac cone in its electronic band structure, signifying its application as a Dirac material.²¹ The presence of Dirac cones in the band structure introduces Dirac fermions with an ultrahigh Fermi velocity ($3.5 \times 10^6 \text{ ms}^{-1}$).¹⁷ Unfortunately, the nonmagnetic nature of hydrogenated borophene limits its application in the field of spintronics. To overcome the limitation and enhance its applicability in such devices, magnetism needs to be associated with hydrogenated borophene. The doping of 3d transition metal atoms is an effective way to introduce magnetism in 2D materials. It has been reported that magnetism in borophenes (2-Pmmn , β_{12} and χ_3) can be effectively introduced by doping of 3d transition metal atoms.^{22–25} However, the quest of introducing magnetism in hydrogenated borophenes by transition metal doping has not been established thus far, presenting a significant gap in the current understanding. We aim to address this scientific gap in the present study. Our study aims not only to pioneer the introduction of magnetism in hydrogenated borophenes *via* transition metal doping, but also to explore novel techniques like the Hubbard correction and Curie/Néel's temperature estimation.

In this work, we employ density functional theory (DFT) simulations to gain deep insights into the electronic and magnetic properties of 3d transition metal atom doped hydrogenated borophenes. We present a systematic investigation of the structural stability of hydrogenated borophenes and the effects of transition metal dopants on the electronic and magnetic properties of the doped system with identification of ferromagnetic (FM) and anti-ferro-magnetic (AFM) states. We have also utilized the mean field theory and Heisenberg model to estimate the critical temperature for AFM and FM states.

2. Computational method and details

The present calculations to study the magnetic interaction in transition metal atom doped hydrogenated borophenes have been performed by utilizing Quantum Espresso,²⁶ a first-principles based density functional theory simulation open-source software package, distributed under the GNU license. The electron–electron interaction is described by the generalized gradient approximation (GGA) exchange functional of Perdew–Burke–Ernzerhof (PBE)²⁷ with the Hubbard U correction (GGA+ U) to accurately capture the on-site Coulomb interactions of 3d transition metal electrons. The Hubbard U value has been calculated using the linear-response method implemented in Quantum Espresso. Ultrasoft pseudopotential has been utilized for interaction between ion and core electrons. A vacuum of 20 Å along the z -axis, perpendicular to the monolayer, has been incorporated to nullify the possibility of layer–layer interaction. The plane wave basis set with a cutoff energy of 60 Ry has been set for the wavefunction, and 480 Ry cutoff energy has been set for the charge density. A $10 \times 10 \times 1$ Monkhorst–Pack²⁸ k -point mesh and a total energy convergence threshold equaling 10^{-6} Ry have been utilized for the structure optimization and self-consistent calculations. A $3 \times 3 \times 1$ supercell of hydrogenated borophene

containing 36 atoms has been considered for doping the 3d transition metal (V, Cr, Mn, Fe, Co, Ni) at the site of one hydrogen atom. All structures have been fully optimized until the force on each atom was less than 10^{-4} Ry Bohr⁻¹. To examine the magnetic and electronic properties, spin polarized calculations have been performed for pristine and doped structures. The bonding and antibonding states have been calculated based on the crystal orbital Hamilton population (COHP) by the local orbital basis suite towards electronic structure reconstruction code (LOBSTER)^{29–32} with Quantum Espresso. The Bader charge analysis has been performed to study the charge redistribution in doped and undoped structures.^{33–36} The phonon dispersion relationship has been calculated using the phonopy code, using the finite displacement and supercell approach.³⁷ To study the magnetic interaction among the dopants, the size of supercells having at least one doped transition metal atom has been doubled along the x -axis. Packages such as VESTA³⁸ and XCrySDen³⁹ have been employed for structure visualization and acquisition of density distribution plots.

3. Results and discussion

3.1. Structural stability of hydrogenated borophenes

Fig. 1a depicts the top view of a borophene sheet and Fig. 1b illustrates the side view of borophene having the space group 2-Pmmn . The lattice parameters and the buckling height of an optimized unit cell are $a = 1.618 \text{ Å}$, $b = 2.862 \text{ Å}$ and $h = 0.894 \text{ Å}$ respectively, which nicely match with the earlier reported values.⁴⁰ To examine the electronic properties, the electronic band has been calculated and it has been found that the 2-Pmmn borophene is metallic due to the band crossing the Fermi level as shown in Fig. 1c, while energy differences of 4.29 eV and 9.67 eV along the $Y\text{-}\Gamma$ and $X\text{-}S$ directions respectively make the borophene energetically anisotropic. The schematics of the top view and the side view of the hydrogenated borophene supercell have been depicted in Fig. 1d and e respectively. The structure expands along the x -axis due to hydrogenation, which causes significant changes in the estimated lattice parameter value of ' a ' (1.945 Å) and a mild change in the lattice parameter value of ' b ' (2.813 Å). The calculated results are in good agreement with the previously reported data.⁴⁰ The full hydrogenation of borophene modifies its electronic state from metallic to semimetallic, with a distorted Dirac cone along the $\Gamma\text{-}X$ direction, as shown in Fig. 1f. This can also be understood from the fact that when hydrogen bonds with a boron atom, the electrons from boron are transferred to hydrogen due to higher electronegativity of hydrogen (2.20) as compared to that of boron (2.04). The amount of charge transferred from boron to hydrogen has been estimated to be 1.06 e , by utilizing Bader charge analysis. The charge density difference and electron localized function plot for hydrogenated borophene have been displayed in Fig. 2, which corroborates well with Bader charge analysis. This electron transfer results in the redistribution of electronic states. The highest stability should place E_F precisely at the zero-point of in-plane PDOS, filling all available in-plane bonding states and the antibonding

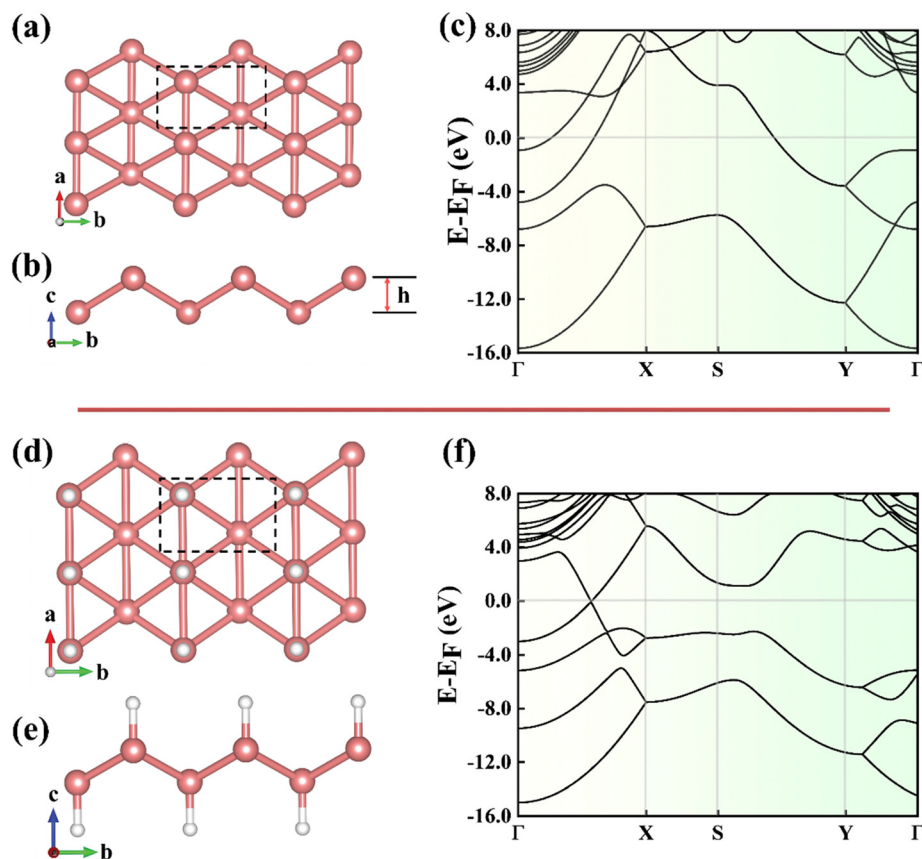


Fig. 1 Structure and band diagram of 2-*Pmmn* borophene and the hydrogenated borophene: (a) top view of borophene with the unit cell outlined in a black dashed line, (b) the side view of borophene, (c) electronic band diagram of a borophene unit cell, (d) top view of the hydrogenated borophene with the unit cell outlined in a black dashed line, (e) side view of the hydrogenated borophene, and (f) electronic band diagram of the hydrogenated borophene unit cell.

states remain unoccupied. The remaining electrons should half-populate the out-of-plane p_z -derived bonding states.⁴¹ Following this guiding principle, the crystal orbital Hamilton population (COHP) along with the total density of states (DOS) and the partial density of states (PDOS) of hydrogenated borophene have been presented in Fig. 3a and b respectively. COHP analysis reveals that the zero-point lies exactly at the Fermi level having occupied bonding states and unoccupied antibonding states. The in-plane PDOS and out-plane PDOS of hydrogenated borophene in Fig. 3b fulfill the condition that in-plane bonding states are completely occupied and antibonding states are unoccupied, which confirms that the hydrogenated borophene is a stable structure. The COHP and PDOS of pristine borophene have been exhibited in Fig. S1 (ESI[†]), which provides a qualitative insight into the redistribution of electronic states before and after hydrogenation. The phonon band structure has been calculated, to further check the stability of the structure, as shown in Fig. 3c, which reveals that there is no negative frequency observed in the phonon band diagram of hydrogenated borophene. A similar analysis for pristine borophene reveals a small imaginary frequency near the Γ point as shown in Fig. S1c (ESI[†]). Thus, the hydrogenation in borophene removes the imaginary frequency making the structure dynamically more stable.

3.2. The structure stability of 3d transition metal doped hydrogenated borophenes

To investigate the magnetic properties, detailed calculations have been performed. The DOS results of borophene and hydrogenated borophene (HB) with spin up and spin down states are shown in ESI[†] Fig. S2 and S3. The symmetric nature of spin up and spin down states enunciates the nonmagnetic nature of borophene and hydrogenated borophene. Our calculations reveal that pristine and the hydrogenated borophenes are nonmagnetic. To investigate the onset of magnetism in the hydrogenated borophene system, we have selectively doped hydrogenated borophenes with 3d transition metals (TM), including V, Cr, Mn, Fe, Co, and Ni, by removing one hydrogen atom from the top layer of the hydrogenated borophene. To accurately assess the 3d state of transition metals in hydrogenated borophene sheets, the Hubbard U value using the linear-response method has been calculated and is displayed in Table 1. The doping of TM causes charge to be transferred between hydrogenated borophene and TM, due to a difference in electronegativity, which affects the stability of the transition metal doped hydrogenated borophene (TM-HB). The Bader charge calculation for doped hydrogenated borophenes is performed to analyze the charge transfer. The calculation revealed the transfer of charge from a transition metal to nearest atoms as shown in Table 1. The

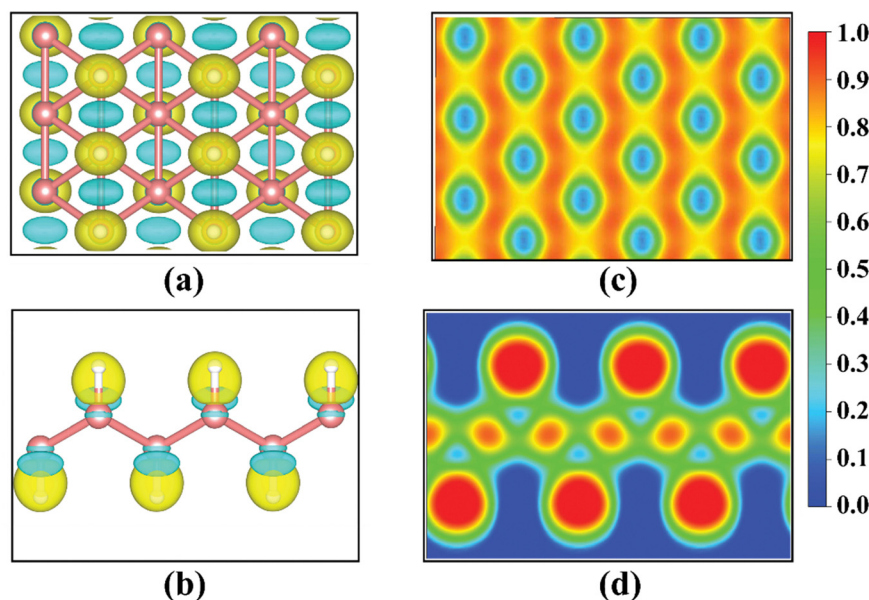


Fig. 2 The plot of charge density difference with an iso-surface value of $0.0144 \text{ e } \text{\AA}^{-3}$ of the hydrogenated borophene: (a) top view and (b) side view with yellow and blue color zones depicting the charge accumulation and depletion respectively. The 2D isosurface of electron localized function (ELF) in the hydrogenated borophene: (c) top view and (d) side view, with the color bar at the right side indicating increasing bond strength and localization with increasing number.

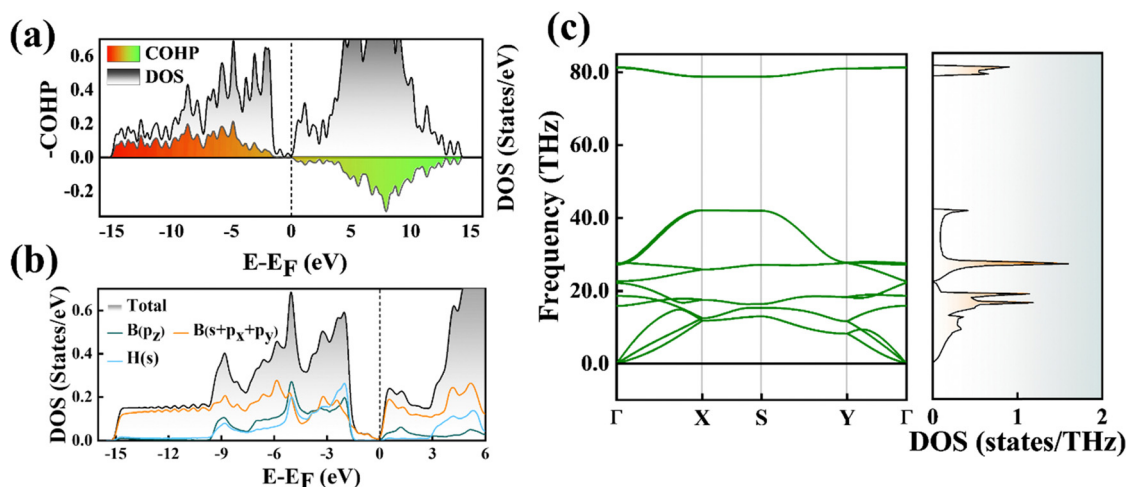


Fig. 3 (a) Crystal orbital Hamilton population (COHP) diagram showing the bonding state (positive, in red color) and the antibonding state (negative, in green color) and the corresponding total DOS, (b) PDOS with in-plane $s + p_x + p_y$, out of plane p_z -state of B and s -state of H atom and (c) phonon band dispersion and phonon density of states of the hydrogenated borophene.

structural stability of TM-HB is established by formation energy (E_f) calculations, utilizing the following equation:⁴²

$$E_f = E_{\text{doped}} + E_{\text{H}} - E_{\text{TM}} - E_{\text{HB}} \quad (1)$$

where E_{doped} is the total energy of TM-HB, and E_{H} and E_{TM} are the total energy of isolated H and TM atoms, respectively. E_{HB} is the total energy of the hydrogenated borophene. The calculated formation energy ranges from 0.78 to -3.64 eV , as displayed in Table 1. The negative formation energy of a few TM doped HB structures, *e.g.* V-HB, Mn-HB and Co-HB structures, indicates the possibility of their potential experimental realization to a reasonable degree. The

positive formation energy of Cr-HB and Co-HB structures suggests that these structures are highly sensitive to the surrounding ambience and might be difficult to achieve in a stable form. Binding energy is another criterion for deciphering the stability of the structures, by revealing the bond strength of a doped atom, which has been obtained by utilizing the following equation:

$$E_b = E_{\text{doped}} - E_{\text{H-vacancy}} - E_{\text{TM}} \quad (2)$$

where E_{doped} is the total energy of TM-HB, $E_{\text{H-vacancy}}$ is the total energy of the hydrogenated borophene with H vacancy, and E_{TM} is the total energy of an isolated transition metal atom. It can be

Table 1 The Hubbard U parameter, the binding energy (E_b), formation energy ($E_{\text{formation}}$), charge transfer and magnetic moment of 3d transition metal atom doped hydrogenated borophene structures

Dopants	U (eV)	E_b (eV)	$E_{\text{formation}}$ (eV)	Charge transfer (e)	Magnetic moment (μ_B)
V	2.72	-2.93	-0.14	0.96	3.49
Cr	5.78	-2.01	0.78	0.89	4.76
Mn	4.51	-3.10	-0.31	0.99	4.00
Fe	4.67	-2.19	0.59	0.60	2.86
Co	4.28	-3.13	-0.34	0.75	1.85
Ni	6.11	-6.44	-3.64	0.22	1.00

established from the negative values of binding energies from Table 1 that all the transition metal atoms chosen in the present case make strong bonds with neighboring atoms in TM-HB structures.

The graphical representations of binding energy, formation energy, charge transfer and magnetic moment induced with respect to various doped transition metals have been illustrated in Fig. 4a to d respectively. Fig. 4a–d illustrate that doping with a lower atomic number dopant results in higher charge transfer and magnetic moment, but the binding energy and formation energy values are not minimum, and the converse is true for doping with higher atomic number transition metal dopants. This gives us a recipe to choose the dopant transition metal by having a balance between stability and magnetic moment induced.

3.3. Magnetic properties

We have performed the spin-polarized calculation for a single impurity transition metal (TM) atom doped $3 \times 3 \times 1$ supercell of the hydrogenated borophene to investigate the magnetic properties. The presence of unpaired electrons after doping plays a decisive role in quantifying the magnetic moment of the system. Our calculations reveal that the Cr-doped system has a

maximum magnetic moment of $4.76\mu_B$. However, the single isolated Cr atom shows magnetization of $6\mu_B$. The change in magnetization value is due to the transfer of charge ($0.89 e$) from Cr to the hydrogenated borophene, which is confirmed by Bader charge analysis as shown in Fig. 4a. The magnetic moment and relative charge transfer for other transition metal doped structures are presented in Table 1. The spin density difference ($\rho_{\text{up}} - \rho_{\text{down}}$) isosurface representation of the transition metal doped structure is shown in Fig. 5 and it is found that the spin density is mostly distributed around the doped atom, which provides an effective substantiation that the existence of the doped transition metal atom is mostly responsible for the onset of magnetization. The shape of the spin density isosurface is uniform around the dopants V, Cr and Mn, as shown in Fig. 5a–c, while Fig. 5d–f elucidates that the spin density is not uniformly distributed around the dopant atoms Fe, Co and Ni. The possible reason for this contrasting behavior for higher atomic number dopants (Fe, Co, Ni) could be the enhanced interaction of d-electron with the hydrogenated borophene's states with d-orbital of doped transition metal atoms. The electronic occupancy of the doped atoms V ($3d^3, 4s^2$), Cr ($3d^5, 4s^1$), Mn ($3d^5, 4s^2$), Fe ($3d^6, 4s^2$), Co ($3d^7, 4s^2$) and Ni ($3d^8, 4s^2$) explains that the majority of the d-orbital is filled by spin up electrons for V, Cr and Mn, which uniformly fills the spin up d_{xy} and $d_{x^2-y^2}$ states, leaving the spin down d_{xy} and $d_{x^2-y^2}$ states mostly unoccupied, causing isotropic spin polarization. While electrons start filling the spin down state of the d-orbitals for Fe, Co and Ni, particularly the d_{xy} state more than $d_{x^2-y^2}$, leading to anisotropic spin polarization in the system, as shown in Table S1 (ESI†).

We have calculated the partial density of states to delve deeper into the magnetic properties of transition metal doped hydrogenated borophenes, as shown in Fig. 6a–f. An asymmetry

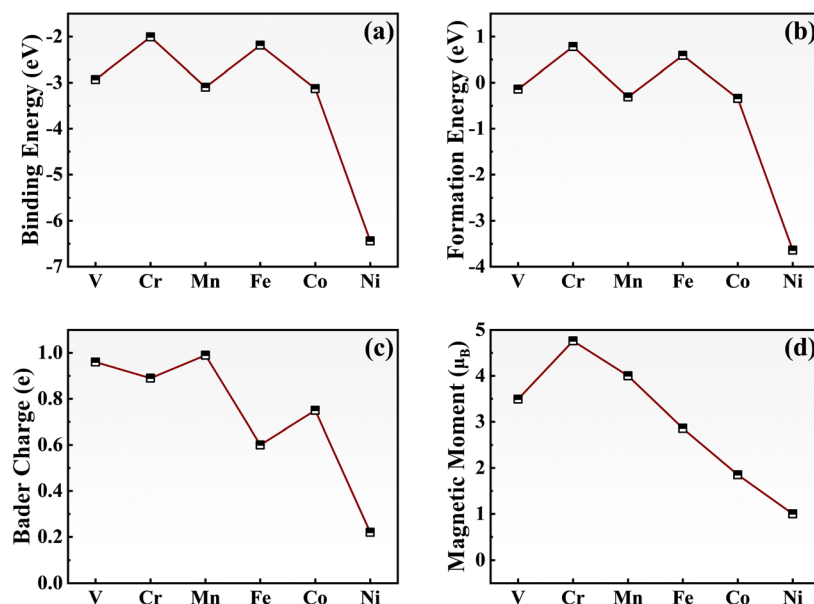


Fig. 4 (a) The binding energy of a 3d transition metal atom doped in the hydrogenated borophene, (b) formation energy of the doped structure, (c) charge transfer from the doped atom to the structure and (d) the magnetic moment of the transition metal atom doped structure.

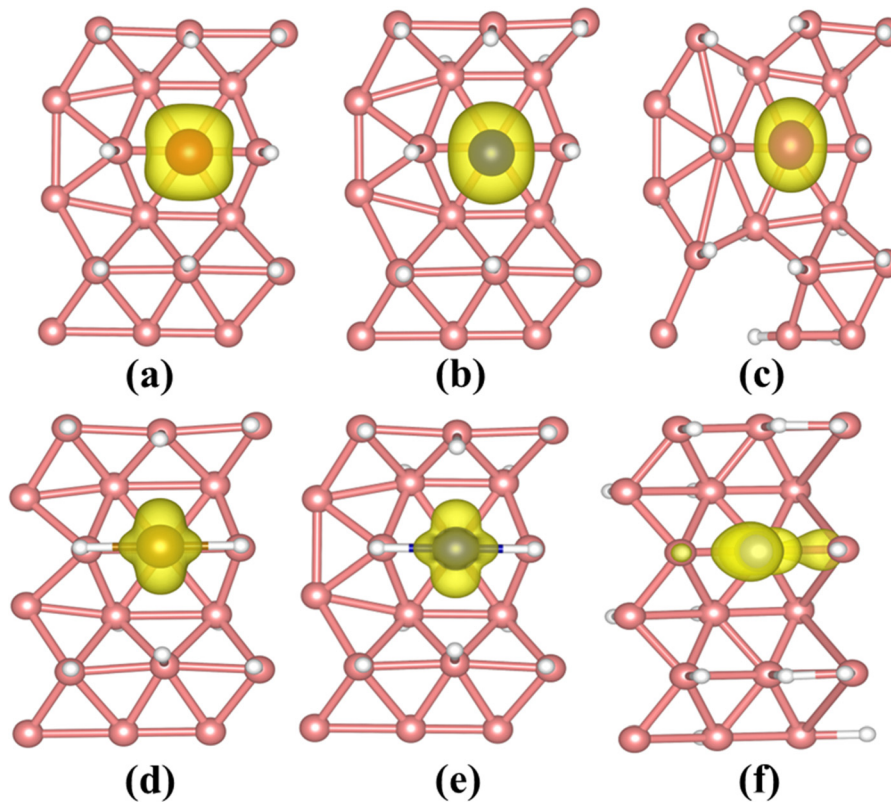


Fig. 5 Spin density ($\rho_{\text{up}} - \rho_{\text{down}}$) isosurface for (a) V-, (b) Cr-, (c) Mn-, (d) Fe-, (e) Co-, and (f) Ni-doped hydrogenated borophene.

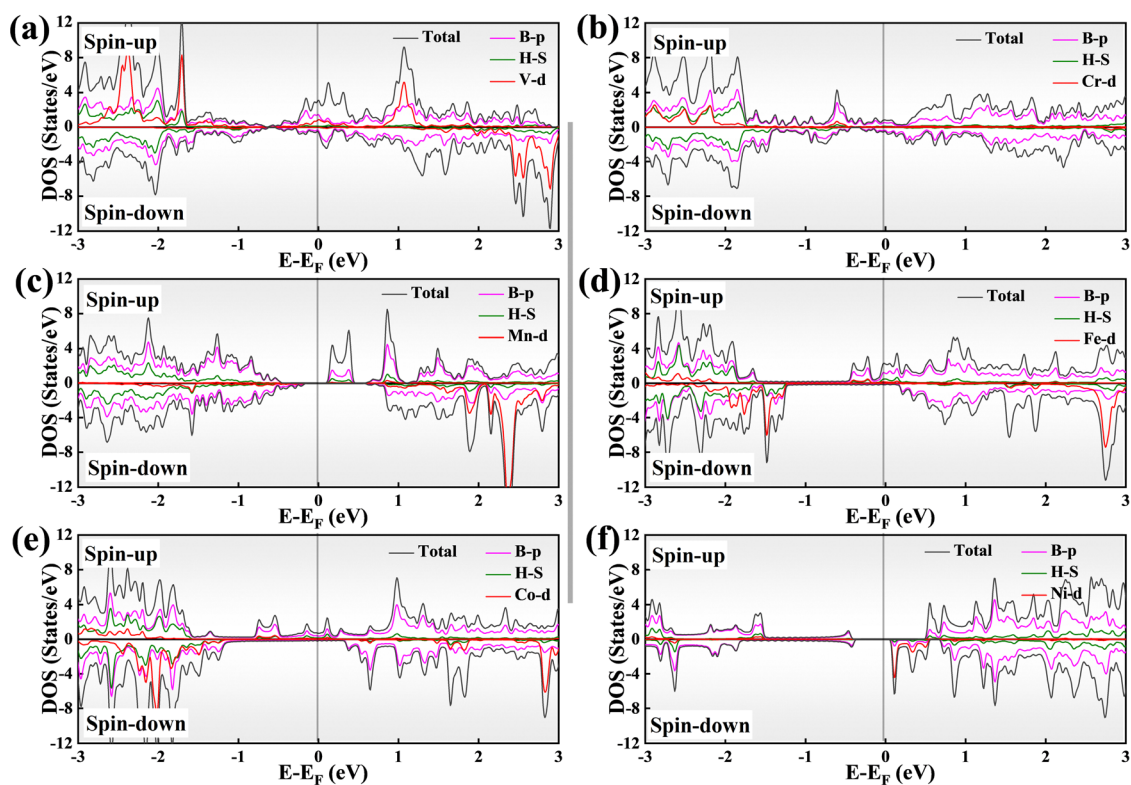


Fig. 6 Partial density of states (PDOS) of (a) V, (b) Cr, (c) Mn, (d) Fe, (e) Co and (f) Ni doped hydrogenated borophene.

between the density of states of spin-up and spin-down states is a strong indicator of the material's magnetic behavior. We can observe from Fig. 6 that a striking asymmetry exists in some of the dopant's cases and the asymmetry is maximum in Fig. 6b for the Cr doped HB structure, which results in a maximum magnetic moment of $4.76\mu_B$ as per Table 1. The spin density states of B(p-state), H(s-state) and TM's(d-state) have been considered in PDOS to understand the orbital interactions responsible for magnetism. The 2p-orbitals of boron and the 3d-orbitals of the TM atom primarily contribute towards the total spin state of the doped structure. This p-d orbital interaction results in a stable magnetization in the transition metal doped hydrogenated borophene (TM-HB). The transition metal atom's 3d orbitals are significantly more spin polarized than boron's 2p orbitals. Thus, transition metal doping in hydrogenated borophenes results in the localized magnetization around the TM atom.

The band structure calculation further helps us to understand the electronic properties of TM-HB, in addition to magnetic property analysis. The electronic band structure of spin up and spin down states has been illustrated in Fig. 7a-f. The undoped hydrogenated borophene exhibits an in-plane Dirac cone in its electronic band diagram with Dirac point at the Fermi level, as shown in Fig. 1f. The spin-up and spin-down states exhibit different electronic configurations, after doping with different transition metal atoms as exhibited in Fig. 7a-f. It can also be observed from these figures that the Dirac point shifts toward the valence band, for the V, Cr, Fe and Co doped HB structures, due to excess electrons in the system after transition metal atom doping. This causes extra population on the Fermi level and the system exhibits metallic behavior, except for the Mn and Ni doped structures, which corroborates well with the density of state results depicted in Fig. 6a-f. In the

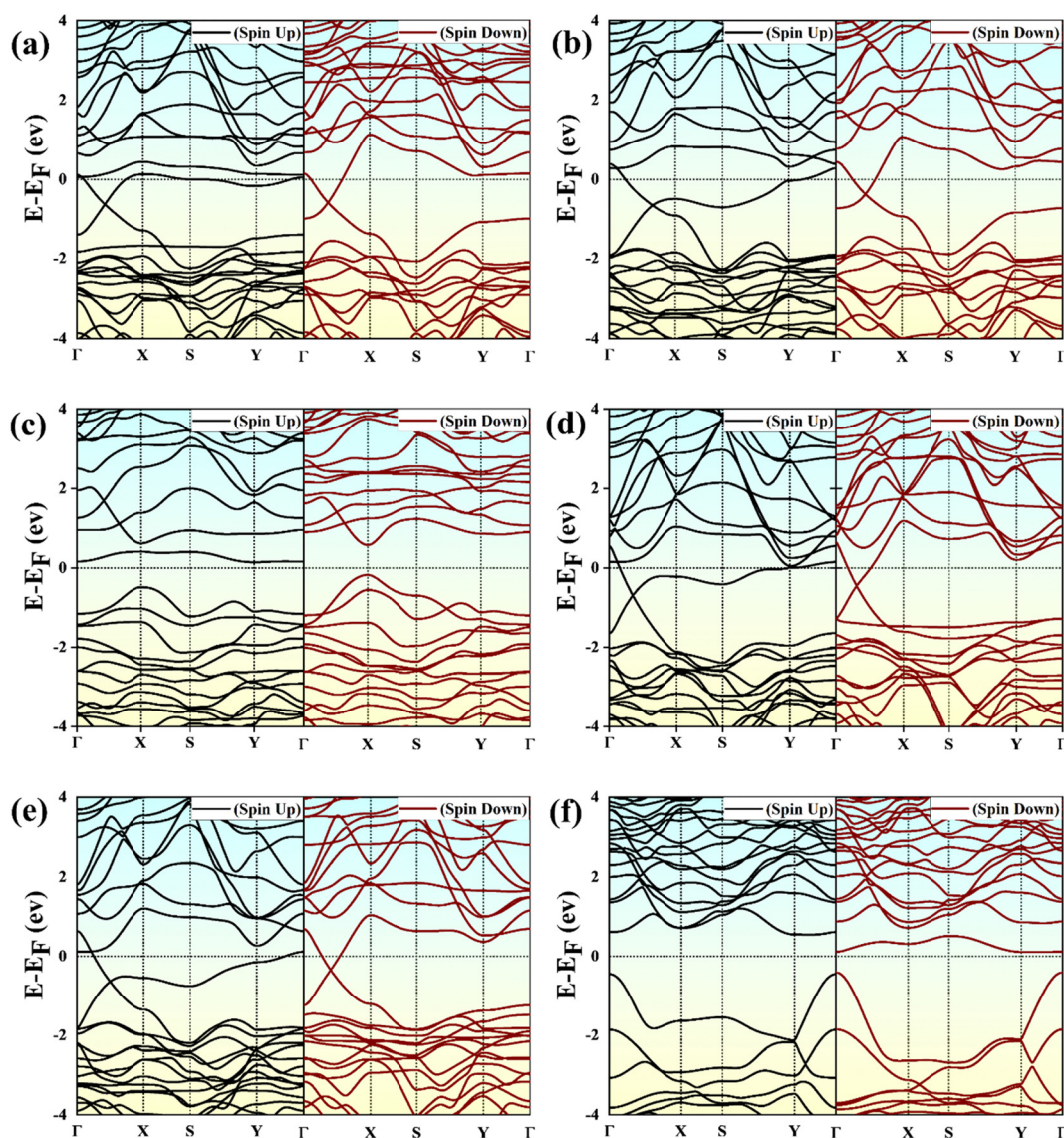


Fig. 7 The band diagram of transition metal doped hydrogenated borophene for (a) V-doped, (b) Cr-doped, (c) Mn-doped, (d) Fe-doped, (e) Co-doped and (f) Ni-doped structures.

band diagram for an Mn doped system shown in Fig. 7c, the valence band maxima lie at the X-point and the conduction band minima lie at the Y-point, which makes it semiconducting with an indirect band gap of 0.62 eV for the spin up state and a direct band gap of 0.75 eV for the spin down state, while the Ni doped system shows a direct band gap of 1.06 eV for the spin up state and 0.52 eV for the spin down state which is also evident from Fig. 6c as well, where the band gap is clearly revealed. The semiconducting behavior of these doped structures in the case of Mn and Ni doping could be pinned upon the localization of d-electrons and the interaction between the d-orbital of dopant and the p orbital of boron atoms. It has been exhibited in the partial density of states of the Mn and Ni doped structure in Fig. 6c and f that the d-orbital electronic states are not distributed in all energy ranges and are confined for some energy by exhibiting a high DOS peak. A high peak in the d-orbital of doped transition metals suggests that the electron is localized around the dopant resulting in a semiconducting state. The electronic configuration of each dopant can also play a critical role in modulating the electronic and magnetic properties. The doping of Mn ($3d^5$, $4s^2$) and Ni ($3d^8$, $4s^2$) induces significant structural changes, as compared to doping by V ($3d^3$, $4s^2$), Cr ($3d^5$, $4s^1$), Fe ($3d^6$, $4s^2$) and Co ($3d^7$, $4s^2$) as demonstrated in Fig. 5. The structural changes and the reduced symmetry can break band crossing and open up the gap in Dirac nodes.⁴³ The variation in band gap with asymmetric band nature makes it useful for spintronic applications.

3.4. Magnetic exchange interaction

We doubled the supercell size to further analyze the magnetic interaction between two transition metal atoms. Antiferromagnetic (AFM) and ferromagnetic (FM) states are assigned to the system by arranging the spin of transition metal atoms antiparallel and parallel. The exchange energy between the two TM atoms is important for their impact on the magnetic properties of the system. The ground state energies of the AFM state (E_{AFM}) and FM state (E_{FM}) have been estimated to assess the exchange energy (ΔE) by using the following formula:

$$\Delta E = E_{\text{FM}} - E_{\text{AFM}} \quad (3)$$

The negative value of exchange energy enunciates that the magnetic state is ferromagnetic, while a positive value of ΔE indicates the antiferromagnetic state. It is found that Cr-, Mn- and Co-doped hydrogenated borophene structures exhibit ferromagnetic coupling, while V- and Fe-doped hydrogenated

borophene structures exhibit antiferromagnetic coupling as displayed in Table 2.

The nonmagnetic ground states of all doped structures have also been calculated (Table 2) and it is revealed that Ni doped hydrogenated borophene is nonmagnetic in its ground state. The mechanisms such as RKKY interactions, p-d exchange, and double exchange could be responsible for the ferromagnetic interaction and the superexchange is responsible for the antiferromagnetic interaction.

3.5. Critical temperature

The magnetic ordering of the system can be FM or AFM as indicated by the value of exchange energy ΔE . According to the mean field theory and the Heisenberg model, the Curie temperature (T_C) for the FM system and the Néel temperature (T_N) for the AFM system are proportional to the exchange energy. To estimate the Curie/Néel temperature, the following relationship has been utilized:^{44,45}

$$T = -\frac{2}{3Nk_B}\Delta E \quad (4)$$

where N is the number of transition metal atoms, k_B is the Boltzmann constant and ΔE is the exchange energy for the energy difference of FM and AFM configurations. The calculated values of critical temperatures are shown in Table 2. The highest Curie temperature (261.9 K) corresponds to the highest value of exchange energy, which is -67.725 meV in the case of the Co doped HB structure, and the negative value of exchange energy suggests that the Co doped structure exhibits ferromagnetic interaction. The highest Néel temperature for the AFM state is 51.096 K, which corresponds to a positive exchange energy (13.209 eV) for the Fe doped HB structure.

3.6. Comparison with other borophene allotropes

Within the context of novel 2D materials, the magnetic properties of the few most investigated transition metal doped 2D borophene allotropes are given for reference in Table 3. In the work of Li *et al.*²², it has been revealed that Cr adsorbed 2-*Pmmn* borophene exhibits a maximum magnetization of $1.43\mu_B$, which is significantly smaller than the isolated magnetic moment of Cr atom ($\sim 6\mu_B$). In the current work, the maximum magnetization has been found for the Cr-doped HB structure, which is $4.76\mu_B$ as displayed in Fig. 4d. This further illustrates that the Cr doped hydrogenated borophene significantly preserves the magnetic moment of the Cr atom in comparison to that reported by Li *et al.*²²

Hydrogenation not only increases the magnetic strength of the 2-*Pmmn* borophene, but also controls the spin-down and spin-up electronic state as displayed in Fig. 7a-f. The state near the Fermi level of transition metal doped hydrogenated borophene is less populated as compared to other borophene allotropes. Ye *et al.*²⁵ in their work have reported a nearly equal magnetization value of $4.53\mu_B$, for Fe atom doped β_{12} borophene; however, both the spin states are highly populated near the Fermi level. Soulen Jr. *et al.*⁴⁸ have suggested that the quantitative analysis of the degree of polarization in a material

Table 2 Exchange energy ΔE , magnetic coupling, and critical temperature for doped hydrogenated borophenes

Dopant	ΔE (meV)	Coupling	T (K)
V	8.650	AFM	-33.461
Cr	-1.515	FM	5.862
Mn	-2.145	FM	8.297
Fe	13.209	AFM	-51.096
Co	-67.725	FM	261.972
Ni	—	NM	—

Table 3 Magnetic moment of a few widely studied transition metal doped 2D materials

	Magnetic moment (μ_B)	Coupling	Mechanism
Hydrogenated borophene	4.76	FM	Cr-doped (present study)
2- <i>Pmmn</i>	1.43	FM	Cr-adsorption ⁴⁶
	0.73	—	Co-adsorption ²³
	3.31	—	Mn-H ₂ S adsorption ⁴⁷
β_{12}	4.53	FM	Fe-doped ²⁵
χ_3	2.86	FM/AFM	Mn-doped ²⁴

Table 4 Degree of spin polarization with spin up and spin down states at the Fermi level for transition metal doped hydrogenated borophenes. The negative sign in the degree of spin polarization shows that down spin has high states at the Fermi level (E_F)

Dopant	$N_{\uparrow}(E_F)$	$N_{\downarrow}(E_F)$	P (%)
V	3.080	0.693	63.265
Cr	0.807	0.549	19.026
Mn	0.0	0.0	—
Fe	2.190	0.181	84.732
Co	0.672	0.154	62.711
Ni	0.0	0.0	—

at the Fermi level (E_F) can be estimated by utilizing the following formula:

$$P = \frac{N_{\uparrow}(E_F) - N_{\downarrow}(E_F)}{N_{\uparrow}(E_F) + N_{\downarrow}(E_F)} \times 100\% \quad (5)$$

where $N_{\uparrow}(E_F)$ and $N_{\downarrow}(E_F)$ symbolize the density of states for spin up and spin down states at the Fermi level. Utilizing this approach, the degree of polarization (P) for transition metal doped hydrogenated borophenes has been calculated and is depicted in Table 4. It can be concluded from Table 4 that the Fe-doped hydrogenated borophene has highly spin polarized states (84.732%). However, V- and Co-doped structures show a relatively minimum degree of spin polarization. The localized magnetization with a high degree of spin polarization makes the hydrogenated borophene a prominent 2D material for spintronic based devices.

Doping of 3d transition metal atoms at the hydrogen site results in fascinating changes in the electronic and magnetic properties. The Cr-doped hydrogenated borophene exhibits a maximum magnetization of $4.76\mu_B$ with the ferromagnetic ground state as shown in Table 2. In this study, we have fixed the doping site and have varied the doped atom (3d transition metals) selectively. Doping of transition metals at other sites like boron may result in different electronic and magnetic properties. The effect of transition metal atom doping at different sites with variable doping concentrations still remains an open area to explore, which may reveal additional novel properties and their application.

4. Conclusions

In this study, density functional theory brings to light the structural, electronic, and magnetic properties of 3d transition metal doped hydrogenated borophenes. Novel approaches,

such as the mean field theory and the Heisenberg model, have been utilized to assess the Curie and Néel temperature for the ferromagnetic and antiferromagnetic states, respectively. The calculated exchange energy confirmed that the Cr-, Mn- and Co-doped hydrogenated borophenes are ferromagnetic while the V- and Fe-doped hydrogenated borophenes are antiferromagnetic. The ground state energy calculations of the ferromagnetic, antiferromagnetic, and nonmagnetic states revealed that Ni is nonmagnetic in its ground state. The maximum magnetic moment of $4.76\mu_B$ is found for the Cr-doped structure while the highest Curie temperature of 261.972 K is estimated for the Co-doped structure. The band structure calculation revealed that the Mn and Ni doped system is semiconducting with different band gaps for spin up and spin down states. The potential features such as ultrahigh Fermi velocity of hydrogenated borophenes, localized magnetization and different band gaps for spin-up and spin-down channels point that the transition metal doped hydrogenated borophenes have the potential to be utilized in future spintronic device applications such as novel quantum materials, spin filtering and data storage, etc.

Author contributions

The manuscript was written through the contributions of all authors. All authors have given approval to the final version of the manuscript.

Data availability

The data supporting this article have been included as part of the ESI.†

Conflicts of interest

There are no conflicts to declare.

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